

Code-Layout, Readability and Reusability

Date	1 July 2026
Team ID	XXXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the developed codebase for the flood prediction system in terms of structure, readability, and reusability. The project follows modular programming practices with proper naming conventions, organized file structure, reusable functions, and well-documented code. These practices improve maintainability and support future enhancements.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and proper code formatting maintained throughout code	Yes	Proper indentation followed
2	Proper File Structure	Files and folders are logically organized by functionality	Yes	Organized into modules
3	Meaningful Variable Names	Variables clearly represent their purpose	Yes	Clear variable naming used
4	Function / Method Names	Functions are descriptively named based on tasks	Yes	Easy to understand functions
5	Code Comments	Inline comments explain logic and processing steps	Yes	Comments added where needed
6	Modular Design	Code divided into reusable modules/functions	Yes	Separate preprocessing and prediction modules
7	No Redundant Code	Duplicate or unnecessary code removed	Yes	Optimized code structure
8	Error Handling	Exceptions and errors handled carefully	Partial	Basic error handling implemented

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas	Dataset Cleaning	High
2	Feature Selection Module	Python	Input Parameter Selection	High
3	Model Training Module	Scikit-learn	ML Model Training	High
4	Prediction Module	Python	Flood Prediction Output	High
5	Visualization Module	Matplotlib	Result Graph Generation	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Code is well organized and modular
Readability	5	Variable names and logic are easy to understand
Reusability	4	Modules can be reused for future prediction systems
Documentation / Comments	4	Sufficient comments added for understanding
Overall Score	4.5/5	Good coding standards followed